IE N5 TRANSISTOR TUT 01

1.

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 The following values of a common emitter amplifier with silicon transistor are known:

$^{+}V_{CC}$ = 30 V β = 300 V_{BE} = 0,7 V R_C = 4,7 kΩ

R_{B1} = 35,541 kΩ R_{B2} = 5 kΩ R_E = 2,2 kΩ

Calculate the values of V_E, I_C, I_B, V_{CE} and V_B of the amplifier. (11)

3.2 The following values of a common-base amplifier are known:

h_{ib} = 40 Ω h_{ob} = 0,4 µA/V h_{rb} = 200 x 10^{-6} h_{fb} = -0,98

R_C = 5 kΩ Z_L = 4 kΩ R_S = 0 R_E = 1 kΩ

Calculate by means of the PRECISE method:

3.2.1 R_L (3)

3.2.2 Z_1 (2)

3.2.3 Z_i (2)
[18]

2.

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 Briefly explain the function of each of the following components in a common emitter amplifier:

3.1.1 Emitter resistor (R_E)

3.1.2 Emitter capacitor (C_E)

(2 × 1) (2)

3.2 The following values of a common emitter are known:

$I_{C(MAX)}$ = 12 mA; R_E = 150 Ω; R_{B1} = 17 kΩ; R_{B2} = 3 kΩ

Calculate each of the following:

3.2.1 $+V_{CC}$

3.2.2 β

3.2.3 V_B

(3 × 3) (9)

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3.3 The following information pertains to a fixed-biased voltage amplifier:

h_{ie} = 1 300 Ω; h_{oe} = 20 µA/V; R_L = 10 kΩ

$h_{re} = 2 \times 10^{-4}$; h_{fe} = 250; R_C = 240 Ω

Calculate by means of the precise method:

3.3.1 The current gain A_i (not necessary for the current-divider rule) (4)

3.3.2 The voltage gain A_V (3)

[18]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 Name THREE factors which cause a variation in the collector current of a transistor because of an unstable temperature. (3)

3.2 The following values of a common-emitter amplifier with a silicon transistor is known:

R_{B1} = 15,947 kΩ; R_{B2} = 3 kΩ; R_E = 120 Ω; R_c = 480 Ω; $^+V_{cc}$ = 12 V;
V_{BE} = 0,7 V and β = 250

Calculate the values of V_e, I_c, I_B, V_{ce} and V_B of the amplifiers. (10)

3.3 Calculate the input impedance Z_i and the output impedance Zo of the circuit in QUESTION 3.2 by means of the approximate method if:

h_{ie} = 1,2 kΩ; $h_{re} = 2 \times 10^{-4}$; h_{fe} = 100 and h_{oe} = 20 µA/V (Rs = 0) (5)

[18]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 The following values of a fixed voltage biased amplifier are known:

$^+V_{CC}$ = 13,2 V; V_{CE} = 6,8 V; β = 300;

$f_{(MIN)}$ = 50 Hz; $I_{C(MAX)}$ = 12,2 mA; V_{BE} = 0,68 V;

h_{ie} = 1,3 kΩ; h_{oe} = 20 µA/V; $h_{re} = 2 \times 10^{-4}$; h_{fe} = 40

Calculate the following:

3.1.1 R_C (2)

3.1.2 R_B (3)

3.1.3 A_i (according to precise method; current divider rule may be used) (4)

3.1.4 A_V (according to precise method) (3)

3.2 Use the circuit in QUESTION 3.1 to calculate the input impedance Z_i and output impedance Zo by means of approximate method. (5)

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QUESTION 3: TRANSISTOR AMPLIFIER

3.1 The following values of a voltage-divider biased emitter amplifier are known:

$^{+}V_{CC}$ = 14 V; V_{CE} = 7 V; β = 350;

$f_{(MIN)}$ = 60 Hz; $I_{C(MAX)}$ = 12,5 mA; V_{BE} = 0,7 V;

Calculate each of the following:

3.1.1 R_e (3)

3.1.2 C_e (2)

3.1.3 R_C (2)

3.1.4 R_{b2} (2)

3.1.5 R_{b1} (3)

3.2 Describe each of the following distortions with the aid of neat, labelled sketches:

3.2.1 Nonlinear

3.2.2 Amplitude

3.2.3 Frequency

(3 × 2) (6)

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QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 3.1.1 Indicate, using TWO sketches, how the operating point on the load line of a common emitter amplifier can shift due to temperature changes. (4)

3.1.2 Briefly explain how the shifting of the operating point could be minimised. (3)

3.2 In a voltage divider biased amplifier the values of R_{B1} and R_{B2} are 74 kΩ and 7,4 kΩ respectively.

Calculate the values of thermal components R_E and C_E, if the minimum signal frequency is 50 Hz and β = 200. (4)

3.3 A fixed biased voltage amplifier has the following information:

h_{ie} = 1 kΩ h_{oe} = 25 µA/V h_{re} = 2,5 x 10^{-4}
h_{fe} = 50 R_B = 200 kΩ R_C = 2,2 kΩ

Calculate according to the PRECISION method the following:

3.3.1 Input impedance of the transistor (3)

3.3.2 Voltage gain of the amplifier (2)

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QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 Choose the correct term from those in brackets. Write only the answer next to the question number (3.1.1–3.1.2) in the ANSWER BOOK.

3.1.1 The voltage gain of a common base amplifier is (high/low/less than one).

3.1.2 The current gain of a common base amplifier is (high/low/less than one).

(2 × 1) (2)

3.2 The following values of a fixed biased amplifier are known:

$+V_{CC}$ = 12 V;	I_C = 10 mA;	V_{CE} = 6 V;	Ve = 1,5 V;
β = 200;	Re = 120 Ω;	Vbe = 0,6 V;	$F_{in(min)}$ = 50 Hz;
h_{ie} = 1,3 kΩ;	$h_{re} = 2 \times 10^{-4}$;		
h_{fe} = 60;	h_{oe} = 20 μ A/V;	R_S = 0	

Calculate each of the following:

3.2.1 R_C (2)

3.2.2 R_B (3)

3.2.3 Z_1 (according to the precise method) (2)

3.2.4 A_i (the current divider rule may be used according to the precise method) (3)

3.2.5 Av (according to the precise method) (3)

3.3 Briefly explain how the shifting of the operating point in a common emitter transistor could be minimised. (1)

[16]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 The following values of a fixed-voltage biased amplifier are known:

R_C = 1,2 kΩ
V_{CE} = 5 V
I_B = 26 μA
β = 300
R_B = 525,385 kΩ

Calculate:

3.1.1 The supply voltage (5)

3.1.2 The base emitter voltage (V_{BE}) (3)

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3.2 Use the values of QUESTION 3.1 and determine the following according to the precise method:

3.2.1 The current gain A_i (without current divider rule) (4)

3.2.2 The voltage gain A_V (3)

Given: h_{ie} = 1,7 kΩ; h_{oe} = 6 µA/V; R_L = 6 kΩ

$h_{re} = 1{,}3 \times 10^{-4}$; h_{fe} = 38

3.3 Define the term *distortion* regarding a common emitter amplifier. (2)
[17]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 Explain the term, *stabilisation.* (2)

3.2 The following values of a common-base amplifier are known:

h_{ib} = 28 Ω; h_{fb} = −0,98 $h_{rb} = 5 \times 10^{-4}$; h_{ob} = 0,34µA/V;
Rc = 8,2 kΩ; Re = 1,2 kΩ; R_L = 12 kΩ

Calculate, by using the precision method, the following values:

3.2.1 The current gain, A_i (use current divider rule) (10)

3.2.2 The input impedance, Z_i (2)

3.3 Name THREE factors that can cause distortion in an amplifier. (3)
[17]